Conditional utility and transfer limits of topological asteroseismic descriptors

Gabriel Wendell

Stage 12 working draft

Abstract

We implement a grouped comparison of compact classical and hybrid seismic models, physical Kepler-to-TESS degradation and frozen-method quality screening. The current dataset supports 9 usable development stars; 11 actual Rudy TESS sectors supply a descriptive transfer case. Scientific G12 is validation_pending. No positive population increment, calibrated detection or population transfer claim follows from this implementation snapshot.

1 Question and controls

The relevant practical question is whether a small collection of geometric descriptors improves prediction after conventional seismic estimates and matched controls are included. Path-superlevel H_0 persistence and peak prominence are the same scalar information. Their location descriptor belongs in both models; the baseline additionally includes a conventional graph-ridge score. Solar-like oscillation scaling and observational context follow standard reviews [1, 2]. The hybrid adds a nominal comb score, the excess competing-alias score and a cylindrical geometry descriptor. A gain within a compact regression class would establish conditional operational utility, not independent spectral information. The observations and published reference centers follow the LEGACY studies [4, 5, 3].

2 Population and prediction tasks

The existing 39/13/13 development/internal/final partition is retained, with Rudy outside model fitting. Features are extracted without reference values; published centers enter only supervised targets and evaluation. Missing inputs and pipeline failures retain their cohort rows. Global parameter regression is the primary task. Catastrophic conventional-estimator failure and nominal-alias reliability are binary quality tasks when class support permits. The selected positive-only population cannot calibrate astrophysical signal absence.

3 Models and grouped validation

For quantity q, the target is $y_i = \log(q_i^{\mathrm{ref}}/q_i^{\mathrm{classical}})$. The fitted correction is $\widehat{q}_i = q_i^{\mathrm{classical}} \exp(\beta_0 + x_i^T\beta)$. Ridge parameters minimize

$$\sum_i w_i(y_i - \beta_0 - x_i^T\beta)^2 + \lambda\|\beta\|^2,$$

where each parent star has total weight one. Median imputation and standardization are fitted inside training folds. Three deterministic outer star folds and three inner star folds isolate penalty selection from out-of-sample prediction. The registered grid is $\lambda \in \{1, 10, 100\}$. Logistic quality models use the same grouped design; an inner-OOF logit layer calibrates probabilities without outer test outcomes. Unsupported class/fold fits are withheld.

4 Uncertainty and practical utility

The primary loss is absolute fractional error. The paired increment is $d_i = |\hat{q}_i^B/q_i^{\mathrm{ref}} - 1| - |\hat{q}_i^H/q_i^{\mathrm{ref}} - 1|$. Prespecified practical margins are 0.01 for $\nu_{\max}$ and 0.005 for $\Delta\nu$. Superiority and two-one-sided equivalence tests receive separate three-test superiority and separate two-regression equivalence corrections. Paired star bootstrap intervals are conditional on the fitted OOF models; full fitting-selection uncertainty is not represented. Empirical intervals use inner OOF residual ranks. They are not exact conformal intervals, and insufficient rank support produces a missing interval. Coverage uses published reference centers rather than known stellar truth. Quality-model Brier scores, fixed-bin calibration and threshold decision curves are reported only with both outcome classes. TESS risk probabilities are withheld under mission shift. A quality flag is not an oscillation detection.

5 Current grouped evidence

Quantity	Stars	Baseline MAE	Hybrid MAE	Improvement
numax	9	0.02456	0.02445	0.00011
dnu	9	0.00768	0.00807	-0.00039

These development OOF measurements are a sparse pilot. Unusable stars are not imputed as successful observations. The source archive includes inherited pipeline failures; their reasons and compatibility repairs are preserved. The registered minimum of 20 development stars is a claim-support guard, not a formal assertion of statistical power. A signed model release must precede the original Stage 11 holdout opening, after which the first archive is applied once.

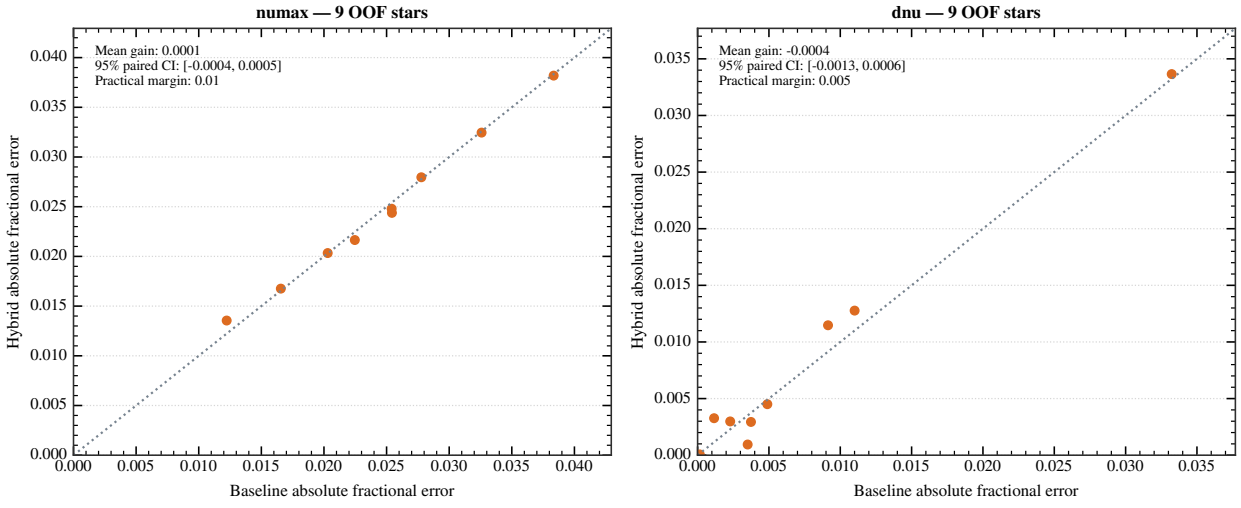


Figure 1: Paired outer-fold errors and conditional bootstrap increments. The sparse development pilot does not support population complementarity.

6 Physical degradation and contribution trajectories

Kepler Q5 samples are integrated as a piecewise-linear flux curve over target exposures. Unsupported exposures and long source gaps are rejected. Cadence, available duration, noise, real sector masks, extra duty loss and contamination are changed along registered trajectories. Seeds depend on parent star, sector, source offset and draw; paired methods and degradation levels share random numbers. Every generated time/flux array and analysis checkpoint is archived. Contamination dilutes variability before instrumental noise is added. Inherited Kepler noise is retained. The unit color-response assumption is a sensitivity; it does not constitute a measured Kepler/TESS passband conversion. The noise model and real-window coverage remain explicit nuisance limitations.

7 Actual Rudy sectors and combinations

The frozen model digest is checked before and during sector processing. All 11 registered SPOC sectors are retained, including quality rejection and nondetection cases. Actual sectors were previously inspected in Stage 10, so this is a reproducible case study rather than new blinded test data. Registered adjacent combinations average PSDs incoherently with exposure weights; they do not coherently stitch stellar phases across years. Candidate estimates are retained separately from accepted measurements. Repeated sectors, offsets and draws still describe one parent star. Conditional degradation quantiles are not population confidence intervals.

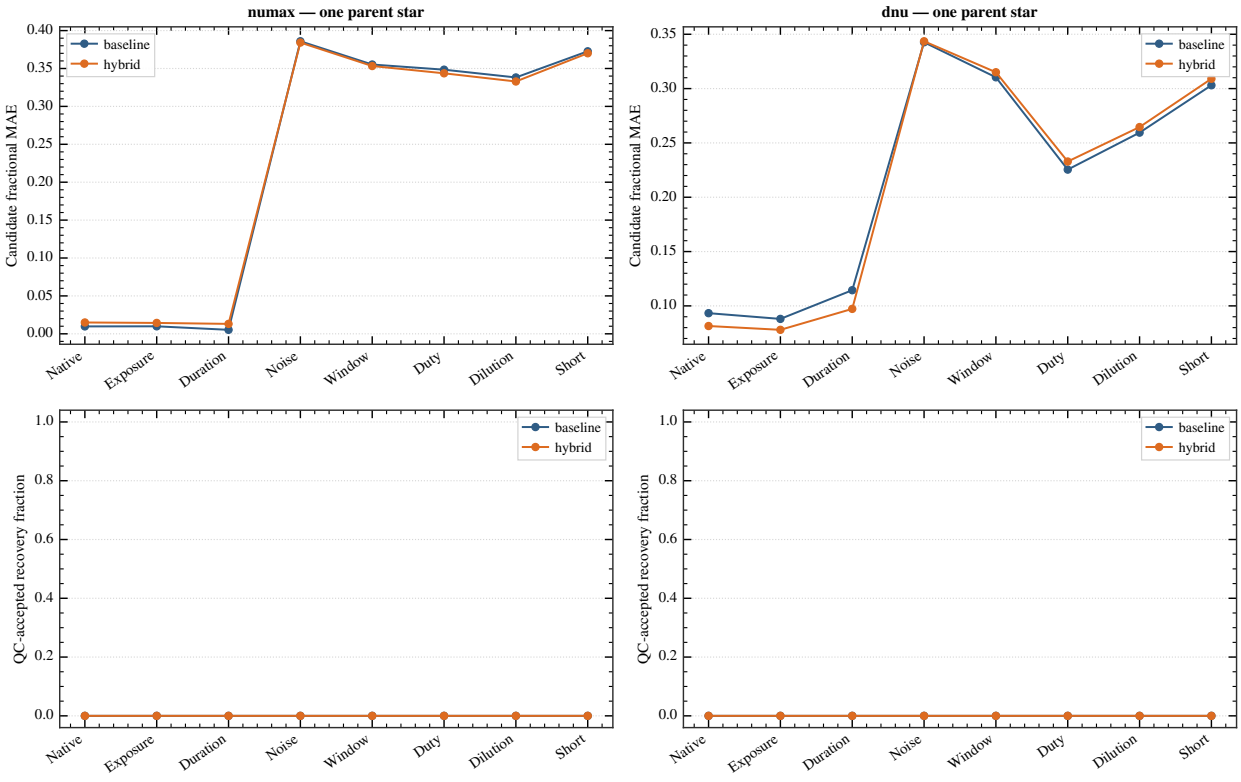


Figure 2: Rudy cumulative degradation trajectories. Rejected candidates remain visible; recoverability additionally requires quality acceptance.

8 Recalibration, scope and counterexamples

A separately authorized layer may estimate a star-weighted median log offset using development simulations from non-Rudy stars. Holdout stars, Rudy and actual test sectors are forbidden from calibration fitting. Features remain fixed, and the calibration layer requires evaluation on disjoint simulation stars. No layer is applied in this snapshot. Missing envelopes, competing aliases, source gaps, out-of-range features, unavailable classes and mission shift are explicit failure domains. The current scope decision stops at Rudy transfer limits; no powered broader TESS sample is established.

9 Reproducibility and claims

The release supplies model JSON, feature/task registries, grouped fold audits, paired results, source-backed figures, degraded light curves, quality reasons, and a model card. Base inference requires NumPy/SciPy only; mission adapters use the existing optional astronomy environment. G7's delay-topology drop and G8's geometry-only interpretation are preserved. Population complementarity requires locked out-of-sample improvement beyond a practical margin or supported failure calibration; positive transfer requires compatible actual observations within uncertainty. Until those conditions are supported, the honest output is a validation-pending pilot and stated transfer limits.

References

[1] William J. Chaplin and Andrea Miglio. Asteroseismology of solar-type and red-giant stars. *Annual Review of Astronomy and Astrophysics*, 51:353–392, 2013.

[2] Rafael A. García and Jérôme Ballot. Asteroseismology of solar-type stars. *Living Reviews in Solar Physics*, 16(1):4, 2019.

[3] M. N. Lund et al. Erratum: Standing on the shoulders of dwarfs. *The Astrophysical Journal*, 850:110, 2017.

[4] Mikkel N. Lund, Víctor Silva Aguirre, Guy R. Davies, et al. Standing on the shoulders of dwarfs: The kepler asteroseismic legacy sample. i. oscillation mode parameters. *The Astrophysical Journal*, 835(2):172, 2017.

[5] Víctor Silva Aguirre, Mikkel N. Lund, H. M. Antia, et al. Standing on the shoulders of dwarfs: The kepler asteroseismic legacy sample. ii. radii, masses, and ages. *The Astrophysical Journal*, 835(2):173, 2017.